

1 EXECUTIVE SUMMARY



MISSION
Enable Indian consumers and businesses to use UPI globally for payments and remittances while creating a seamless cross-border payment network.



PRODUCT THESIS

UPI International is attempting to solve three major pain points in global payments:



HIGH COST



SLOW SETTLEMENT



FRAGMENTED EXPERIENCE



KEY INSIGHT

Cross-border payments are not just a payments problem; they are fundamentally a trust, security, and compliance problem.

2 ABOUT NPCI INTERNATIONAL

WHAT NPCI INTERNATIONAL OFFERS



UPI International (P2M)



Cross-Border Remittances (P2P)



RuPay Global

TARGET USERS



Indian Travelers



NRIs



Foreign Merchants



Partner Banks



Payment Networks

VALUE PROPOSITION

FOR USERS

- Pay abroad using familiar UPI apps
- Lower transaction fees
- Real-time payments

FOR MERCHANTS

- Access Indian spending power
- Faster settlement
- Lower acceptance costs

3 THE PROBLEM

TODAY'S CROSS-BORDER PAYMENT EXPERIENCE

- ✗ Expensive
- ✗ Slow
- ✗ Fragmented
- ✗ Multiple intermediaries
- ✗ Inconsistent across countries

HIDDEN PROBLEM

- Data crosses multiple jurisdictions
- Security standards vary
- Regulations differ
- Liability ownership becomes unclear



KEY INSIGHT

The weakest participant in the payment chain often determines the overall security of the transaction.

THIS TRANSFORMS THE PROBLEM FROM:

"How do we make payments cheaper?"

"How do we make international payments trustworthy?"

4 MARKET OPPORTUNITY

TAM



350M+



25M+



30M+

SAM

(Users likely to transact internationally)



Frequent Travelers



Students Abroad



Business Travelers



NRIs

SOM

(Proven corridors)



India to Singapore



India to UAE



India to Thailand



India to Nepal



India to France

STRATEGIC OBSERVATION

UPI does not need to win everywhere. Winning high-traffic corridors first creates network effects.

5 COMPETITIVE LANDSCAPE

NETWORK

VISA

Mastercard

SWIFT

Alipay

NPCI INTERNATIONAL

REAL MOAT

Trust + Acceptance

Dispute Infrastructure

Compliance

Ecosystem Strength

Cost + Speed



STRATEGIC INSIGHT

UPI is competing against decades of trust infrastructure, not merely payment rails.

UPI has solved:

- ✓ Cost
- ✓ Speed

UPI must now solve:

- A Trust
- A Security
- A Compliance

6 PERSONAS (ENHANCED)



INTERMITTANT TRAVELLER

GOALS

- Pay seamlessly abroad
- Avoid currency hassles

PAIN POINTS

- Merchant acceptance uncertainty
- Fear of fraud abroad
- Payment failure anxiety
- Poor dispute resolution



NRI

- Send money home quickly

PAIN POINTS

- Transaction reversals
- Compliance delays
- Data privacy concerns
- Unclear dispute ownership



MERCHANT

- Accept Indian customers
- Grow business

PAIN POINTS

- Integration complexity
- Settlement uncertainty
- Fraud liability concerns
- Low demand visibility

7 USER JOURNEY + RISK MAPPING

STEP	USER ACTION	HIDDEN RISK
DISCOVERY	Finds UPI merchant	Fake merchant listing / misleading info
PAYMENT INITIATION	Scans QR code / to pay	Fake QR code / QR tampering
AUTHENTICATION	Enters OTP / Uses biometrics	Credential theft / OTP interception
PROCESSING	Cross-border routing	Weakest partner vulnerability
SETTLEMENT	Merchant receives funds	Reconciliation failure / Settlement delay
DISPUTE	Refund / dispute initiated	Multi-country ownership & resolution complexity

Insight: Every additional participant increases security risk, compliance burden and failure probability.

8 ROOT CAUSE ANALYSIS



LAYER 1 ADOPTION PROBLEMS

- Low merchant acceptance
- Low user awareness



LAYER 2 TRUST PROBLEMS

- Payment failure anxiety
- Refund concerns
- Fraud fears



LAYER 3 PLATFORM PROBLEMS

- Regulatory fragmentation
- Security inconsistencies
- Data localization restrictions



ROOT CAUSE

Lack of Global Trust Infrastructure

9 SOLUTION FRAMEWORK

PILLAR 1: ACCEPTANCE LAYER



- Merchant Expansion (priority countries)
- Local acquiring partnerships
- QR interoperability
- Merchant discovery & map



PILLAR 2: TRUST LAYER

- Real-time Fraud Engine (AI/ML)
- Risk-based Authentication
- Behavioral Biometrics
- Real-time Monitoring & Alerts



PILLAR 3: COMPLIANCE LAYER

- Global Compliance Engine
- Country-specific rule automation
- Data Localization Layer
- Regulatory monitoring & reporting

10 PRIORITIZATION ROADMAP

PHASE 1 (0-6 Months)



MERCHANT ACCEPTANCE
Goal: Increase usable locations

PHASE 2 (6-12 Months)



VISIBILITY & AWARENESS
Goal: Increase discovery & usage

PHASE 3 (12-24 Months)



TRUST INFRASTRUCTURE
Goal: Reduce transaction anxiety

PHASE 4 (24+ Months)



COMPLIANCE AUTOMATION
Goal: Scale across more countries

11 NORTH STAR METRIC



SUCCESSFUL & SECURE CROSS-BORDER TRANSACTIONS

Why this metric?



Balances Adoption



Ensures Reliability



Builds Trust



Drives Sustainable Scale

It captures volume, reliability and trust together.

12 SUCCESS METRICS



GROWTH METRICS

- Monthly Active International Users
- Merchant Acceptance Count
- Transaction Volume (GMV)
- New Country Rollouts



RELIABILITY METRICS

- Transaction Success Rate
- Payment Latency
- Settlement Success Rate
- System Uptime



TRUST METRICS

- Fraud Rate (%)
- Dispute Rate (%)
- Refund Turnaround Time
- User Trust Score (NPS)



COMPLIANCE METRICS

- Regulatory Incidents
- Data Privacy Violations
- Partner Compliance Score
- Audit Pass Rate



North Star is supported by metrics across Growth, Reliability, Trust and Compliance.

13 KEY RISKS & MITIGATION

RISK	IMPACT	MITIGATION
Fraud & Scams	Loss of trust, financial loss	AI fraud engine, risk-based authentication, user education
Regulatory Changes	Compliance penalties, service disruption	Regulatory monitoring, flexible compliance engine
Low Merchant Adoption	Low transaction volumes	Incentives, easy integration, local partnerships
Technical Failures	Payment failures, revenue loss	Resilient architecture, real-time monitoring, failover systems
Data Privacy Breaches	Legal penalties, reputation damage	Data localization, encryption, access controls

14 CONCLUSION

UPI INTERNATIONAL – THE NEXT CHAPTER



- UPI has proven India can build world-class payment rails.
- The next challenge is building the world's most trusted cross-border payment network.

PAYMENT RAILS CAN BE REPLICATED. TRUST INFRASTRUCTURE CANNOT.

Trust + Compliance + Simplicity = Global UPI Leadership